

SAP BASIS ECC CONCEPTS

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T-Codes

User Administration

SU01	User Maintenance (Single User Creation / Lock & Un-Lock Users)
SU01D	User Display
SU10	Mass User Creation (Multiple Users)
EWZ5	Overview of Lock & Un-Lock Users
SUIM	Overview of Users Data
SMLG	Create Log On Load Balance

Client Administration

SCC1	Transport Role / Import Role
SCC3	Copy Logs / Transport Log Analysis
SCC4	Client Creation
SCC5	Client Deletion
SCC7	Client Import
SCC8	Client Export
SCC9	Remote Client Copy
SE37	Lock & Un-Lock Clients
SCCL	Local Client Copy

Database Administration

DB01	Analyse Exclusive Lock Waits
DB02	Analyse Tables and Indexes
DB03	Database Parameters
DB12	DB Backup Monitor
DB13	DBA Planning Calendar
DB14	DB BR Tools Logs
DB15	Data Archiving: Database Tables
DBACOCKPIT	Monitor DB Logs

Background Jobs Administration

SM36	Background Job's Design / Standard Background Job's Design
SM37	Background Job's Monitoring / Background Job's Deletion

Spool Administration

SP00	Spool and related areas
SP01	Print Request Overview
SP02	Display Spool Requests
SPAD	Create Spool Administration Request

Transport Management System

SE01	Transport Organizer (Extended View)
SE03	Transport Organizer Tools
SE06	Set Up Transport Organizer
SE09	Transport Organizer
SE10	Releasing Request
STMS	Create TMS Configuration (Transport Management System)
STMS_IMPORT	TMS Import Queue Task

Alert Monitoring T – Codes

RZ20	CCMS Monitoring
AL01	SAP Alert Monitor
AL02	Database Alert Monitor
AL05	Monitor Current Workload
AL18	Local File System Monitor
AL16	Local Alert Monitor for Operating System

Other Administration T – Codes

AL11	Display SAP Directories
SICK	Installation Check
SM35	Batch Input Monitoring
SM58	Asynchronous RFC Error Log
SM59	Remote Function Call Destination (Display/Maintain)
SM01_CUS	Lock & Un-Lock T-codes
SM02	System Message Send
BD54	Logical System Creation & Deletion
SE38	Background Job's Pause / Client Size
SU24	Relation b/w T-codes & Authorization Objects
SU53	Missing Authorizations
DICO	Delete TMS Configuration
SCUM	Create CUA & Distribution Model
SNRO	Increases no. of TEMSE Requests
PFCG	Profile Creation & Generation (Single, Derived & Composite Roles)
SAINT	SAP Add On Installation Tool
SPAM	Support Package Manager
SM14	Update Program Administration
SM56	Number Range Buffer
SM04	User Activities / User Overview
AL08	Global User List (all instance)
SM63	Display/Maintain Operation Modes
RZ03	Check Status of Server
RZ04	Operational Mode Creation & Deletion
ST03	Perform Workload Analysis
RZ10	Profile Maintenance & Change SAP Static Parameter Value / SAP* Enable
RZ11	Dynamic Parameter Creation

ST01	SAP System Trace
RZ12	RFC Server Group Maintenance
SMGW	Gateway Monitor
ST07	Application Monitor
ST02	Tune Summary (view sap buffer & memory configuration)
ST04	Performance Overview
ST05	SQL Trace Performance Trace (works performed by users on system written in this trace)
SM18	Reorganize Security Audit Log
SM19	Enable Security Audit Log
SM20	Analysis of Security Audit Log
ST11	Monitor the Trace and Log

Daily Monitoring T – Codes

SM50	Work Process Overview for Individual Instances
SM51	Monitoring Work Process for Instance wise
SM12	Monitoring Lock Entries
SM13	Monitoring Update Process
DB02	DB Space Overview
DB12	Monitor DB Backup Logs
SM37	Overview of Background Job
SMQ1	Outbound Queue Monitor
SMQ2	Inbound Queue Monitor
ST22	ABAP Runtime Error Analysis
ST06	Monitoring CPU Utilization
SM21	Monitoring System Logs
SM66	Work Process Overview for System wide (Globally(Multiple Instances))

Other BASIS T - Codes

SLICENSE	SAP License Administration
SNOTE	Adding SAP Notes
I18N	Adding Language
SMLT	Importing Language
SU25	Initial Profile Generator
SGEN	SAP Load Generator
SE12	ABAP Dictionary
SE13	Dictionary: Technical Settings
SBWP	Business Workplace of User
SO00	Create Document and Send Mail
SOST	Status of Mail
SCOT	Node Creation for Mail Configuration
SALE	Create Logical System
SCUA	Create Model View (Central User Administration)
WE05	Check Intermediate Doc (IDocs)
WE20	Partner Profiles
WE21	Ports in IDocs Processing
BD64	Display Distribution Model
SP12	TEMSE Management (Temporary Sequential)

Introduction to SAP BASIS

Basis is a set of programs and tools that act as an interface with Database, Operating system, Communication protocols and business applications (such as FI, CO, MM, etc.). Full form of **BASIS** is "**Business Application Software Integrated solution**".

SAP applications such as FI, CO, PP etc. can run and communicate with each other across different Operating systems and Databases with the help of BASIS.

Nowadays Basis is known as **NetWeaver**.

Alias of BASIS is SAP Application Server Technology and alias of NetWeaver is SAP Web Application Server.

After adding **java stack (the applications which are developed in J2EE, BSP, JSP, etc.) enhanced security standard for business process**. Both ABAP and Java stack can be monitored from one platform. NetWeaver supports standard protocols such as HTTP, SMTP, XML, SOAP, SSO, WEBDAV, WSDL, WMLSSO, SSL, X.509 and Unicode format (**representation of handling text**).

We can say **Basis is the operating system for SAP applications and ABAP**. Basis provides services like communication with the operating system, database communication, memory management, runtime collection of application data, web requests, exchanging business data etc...

Basis supports a number of known operating systems (Unix flavours, Microsoft windows server edition, AS400, z/OS, etc.) and databases (Oracle, DB2, Informix, Maxdb, Microsoft SQL Server, etc.).

As we know BASIS is a set of tools. This tool has the following different functionalities:

- System monitoring and administration tools
- Common monitoring tool CCMS (**Computing Centre Management System**) to monitor alerts of R/3 system from one place.
- Server side scripting in ABAP and JavaScript.
- Use of Business server pages to build online stores and portals.
- Database monitoring and administration utilities
- Resource management like memory, buffer, etc.
- Authorization and profile management tools for user management.
- Internet access control to the system and business objects.
- Transfer modifications in a screen, program, layout from the development to a production system for accuracy purpose by **Transport Management System**.
- Client-server architecture and configuration.
- Graphical User Interface designing for the presentation layer.

SAP Basis consultant's responsibilities:

- SAP Basis is a middleware tool for applications, operating system, and database. SAP Basis consultant should able to do the following tasks: -
- SAP application server monitoring, ABAP dump, and system log analysis.
- Performance tuning
- Database maintenance, Database backup schedule and restore
- R/3, NetWeaver, solution manager installation, etc.
- SAP license maintenance.
- SAP landscape, transport management system installations, etc.
- Client creating, client copying, client deletion, etc.
- Creating user, assigning roles, locking and unlocking users, etc.

- Background jobs scheduling, job monitoring, job deletion, etc.
- Profile and operation mode maintenance
- Applying support patches, upgrading and installing add-ons
- SNOTE applying and removing errors.
- System copy, System refresh, etc.

This is a generic list. There are many other responsibilities that a Basis consultant should shoulder. Every day you learn something new!

ECC stands for SAP ERP Central Component, which is an on-premises enterprise resource planning (ERP) system. It's a key technology in business that integrates data from different areas of a company in real time.

Introduction to ERP

SAP is a market leader in providing ERP (Enterprise Resource and Planning) solutions and services.

Enterprise Resource Planning (ERP) is a software that is built to organizations belonging to different industrial sectors, regardless of their size and strength. Its purpose is to manage any organization/company functions. It is a way to integrate the data and processes of an organization into one single system.

Example - SAP, Oracle, NetSuite, Epicor, Microsoft, Dynamics 365 etc.

The ERP package is designed to support and integrate almost every functional area of a business process such as procurement of goods and services, sale and distribution, finance, accountings, human resource, manufacturing, production planning, logistics & warehouse management.

How to Download & Install SAP GUI for Windows

Before you can configure and use the SAP GUI, you need to download the software from the SAP Marketplace as per steps below

Time to configure your GUI

1.Connection Type: - Custom Application Server (Particular one host)

2. Description: - Name of instance

3. Application Server: - IP address of remote application server

4. Instance number which you can find from os level (Unix) GoTo /usr/sap/Sid/DVEBGMS00 Here instance number = 00

5. System ID: - As per your setting which you have specified during installation time.

SAP Instance & SAP SID

What is an Instance

Sap Instance is a group of resources such as

- Memory
- Work Processes
- Dispatcher
- Gateway

usually for a single application or database server within a SAP R/3 client-server environment.

There are three types of instances:

1. Dialog instance

2. Central Instance

3. Database Instance

```
SAP System= Dialog Instance + Central Instance + Database Instance.
```

For one SAP system, all three instances share the same directory.

- **Dialog Instance:** Dialog instance exists in the application layer. Its purpose is to maintain the load on the server. Dialog instance exists on the different host. If a number of dialog instance increases hardware resources, dispatcher, work processes also increases so that more number of users can login at a time.
- **Central Instance:** Central instance can also work as dialog instance. But the main thing is that it contains Enqueue and message servers. All dialog instances communicate with central instance before requesting database with message server. When an instance is started, the dispatcher process attempts to establish a connection to the message server so that it can announce the services it provides (DIA, BTC, SPO, UPD, etc.). Lock table is managed in central instance by Enqueue service.
- **Database Instance:** As normal database instance accepts requests from central instance to fulfil the user's requests. As lock management system provided by Enqueue server, it will provide service to users.

What is SID

SID is a unique identification code for every R/3 installation (SAP system) consisting of a database server & several application servers. SID stands for SAP System Identification. SAPSID — a three-character code such as C11, PRD, E56, etc.)

Logical System Names:

When data is distributed between different systems, each system within a network has to be clearly identifiable. The "logical system" deals with this issue.

A logical system is an application system in which the applications work together on a common database. In SAP terms, the logical system is a client.

Since the logical system name is used to identify a system uniquely within the network, two systems cannot have the same name if they are connected to each other as BW systems or as source systems, or if there are plans to connect them in any way.

Example for production system logical system name might be:

SID – PBG

SID Description - P=Production(type), B=BW(component), G=Germany. (plant name)

Logical System name

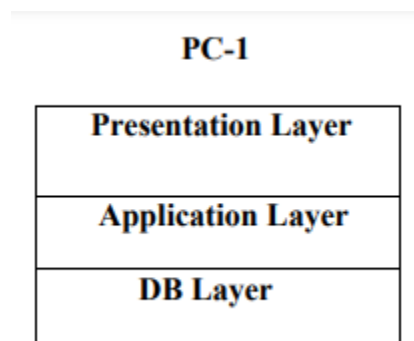
PBGCLNT100.This form is easy to understand.

R/1, R/2, R/3 Architecture

R/1 System

R- Real Time

1 Tire



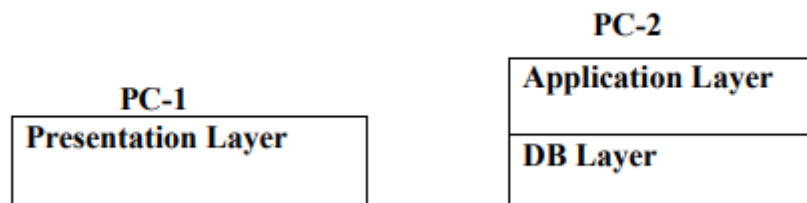
Note: Here in Application Layer Work Process Are going on.

Drawback: If we Lost Power in Presentation Layer we can't able to Login Remaining Systems.

R/2 System

R- Real Time

2- 2 Tire



Drawback: If we Lost Power in Presentation Layer we can't able to Login Remaining 2 Layers.

There is No Q-Mechanism

R/3 System

SAP R/3 is a 3 tier architecture consisting of 3 layers

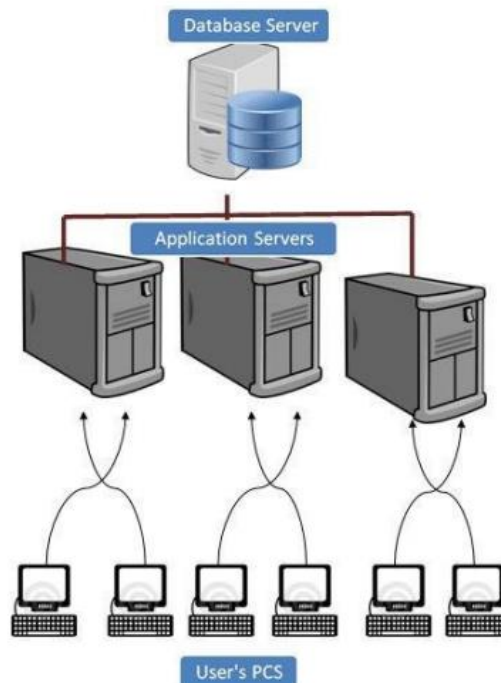
1. Presentation

2. Application

3. Database

In simple words, it's a client server architecture.

- **R** signifies Real-time system.
- **3** represents - 3-tier architecture.



User's PC:

Users can access SAP system in two ways:

1. Through SAP GUI
2. Through Web Browser

It's called front-end. Only the front-end is installed in the user's PC not the application/database servers.

Front-end takes the user's requests to database server and application servers.

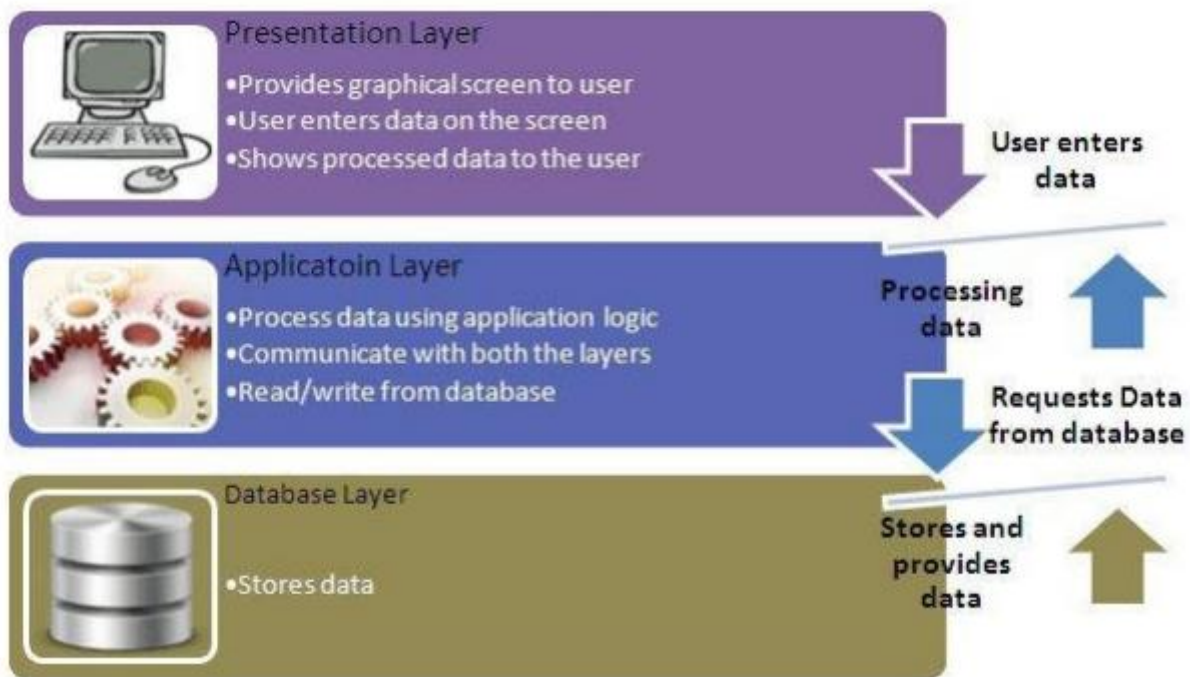
Application Servers: Application server is built to process business-logic. This workload is distributed among multiple application servers. With multiple application servers, the user can get the output more quickly.

Application server exists at a remote a location as compared to the location of the user PC.

Database Server: Database server stores and retrieves data as per SQL queries generated by ABAP and Java applications.

Database and Application may exist on the same or different physical location.

Understanding different SAP layers



Presentation Layer

The **Presentation Layer** contains the software components that make up the SAP GUI (graphical user interface). This layer is the interface between the R/3 System and its users. The R/3 System uses the SAP GUI to provide an intuitive graphical user interface for entering and displaying data.

The presentation layer sends the user's input to the application server, and receives data for display from it. While a SAP GUI component is running, it remains linked to a user's terminal session in the R/3 System.

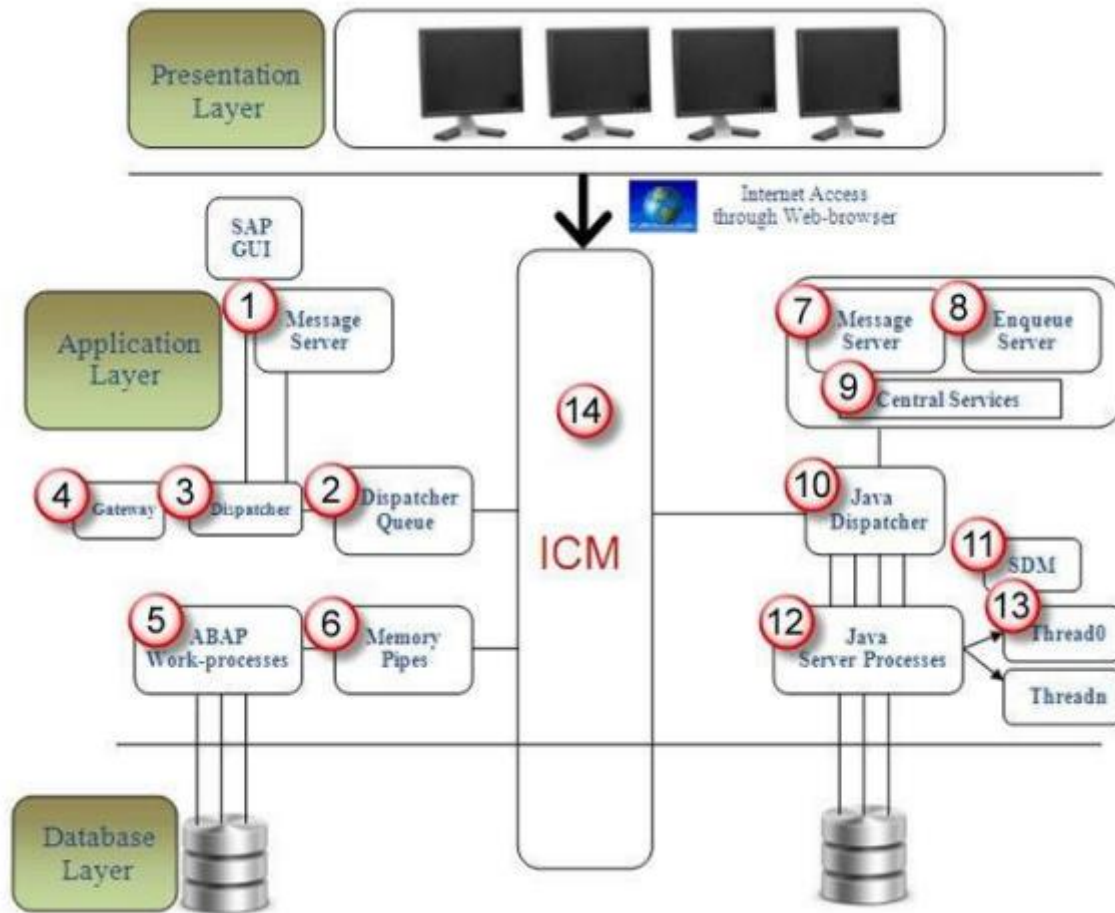
Application Layer

The **Application Layer** consists of one or more application servers and a message server. Each application server contains a set of services used to run the R/3 System. Theoretically, you only need one application server to run an R/3 System. In practice, the services are distributed across more than one application server. The message server is responsible for communication between the application servers. It passes requests from one application server to another within the system. It also contains information about application server groups and the current load balancing within them. It uses this information to assign an appropriate server when a user logs onto the system.

Database Layer

The **Database Layer** consists of a central database system containing all of the data in the R/3 System. The database system has two components - the database management system (DBMS), and the database itself. SAP has manufactured its own database named Hana but is compatible with all major databases such as Oracle. All R/3 data is stored in the database. For example, the database contains the control and customizing data that determine how your R/3 System runs. It also contains the program code for your applications. Applications consist of program code, screen definitions, menus, function modules, and various other components. These are stored in a special section of the database called the R/3 Repository, and are accordingly called repository objects. R/3 repository, objects are used in ABAP workbench.

Understanding the components of SAP R/3 3-tier Architecture:



ABAP+Java System Architecture

- 1. Message Server:** It handles communication between distributed Dispatchers in ABAP system.
- 2. Dispatcher Queue:** Various work process types are stored in this queue.
- 3. Dispatcher:** It distributes requests to the work processes.
- 4. Gateway:** It enables communication between SAP system and between SAP system and external systems.
- 5. ABAP-Work processes:** It separately executes dialog steps in R/3 applications.

Types of work processes are given as below: -

Dialog	• Responsible for dialog process
Update	• Responsible for update
Update2	• Responsible for less time critical update
Background	• Responsible for background job
Spool	• Responsible for output requests
Enqueue	• Responsible for locks

6. Memory-pipes: It enables communication between ICM and ABAP work processes.

7. Message Server: It handles java dispatchers and server processes. It enables communication within java runtime environment.

8. Enqueue Server: It handles logical locks that are set by the executed Java application program in a server process.

9. Central Services: Java cluster requires a special instance of the central services for managing locks and transmitting messages and data. Java cluster is a set of processes that work together to build the reliable system. Instance is group of resources such as memory, work processes and so on.

10. Java Dispatcher: It receives the client requests and forwards to the server process.

11. SDM: Software Deployment Manager is used to install J2EE components.

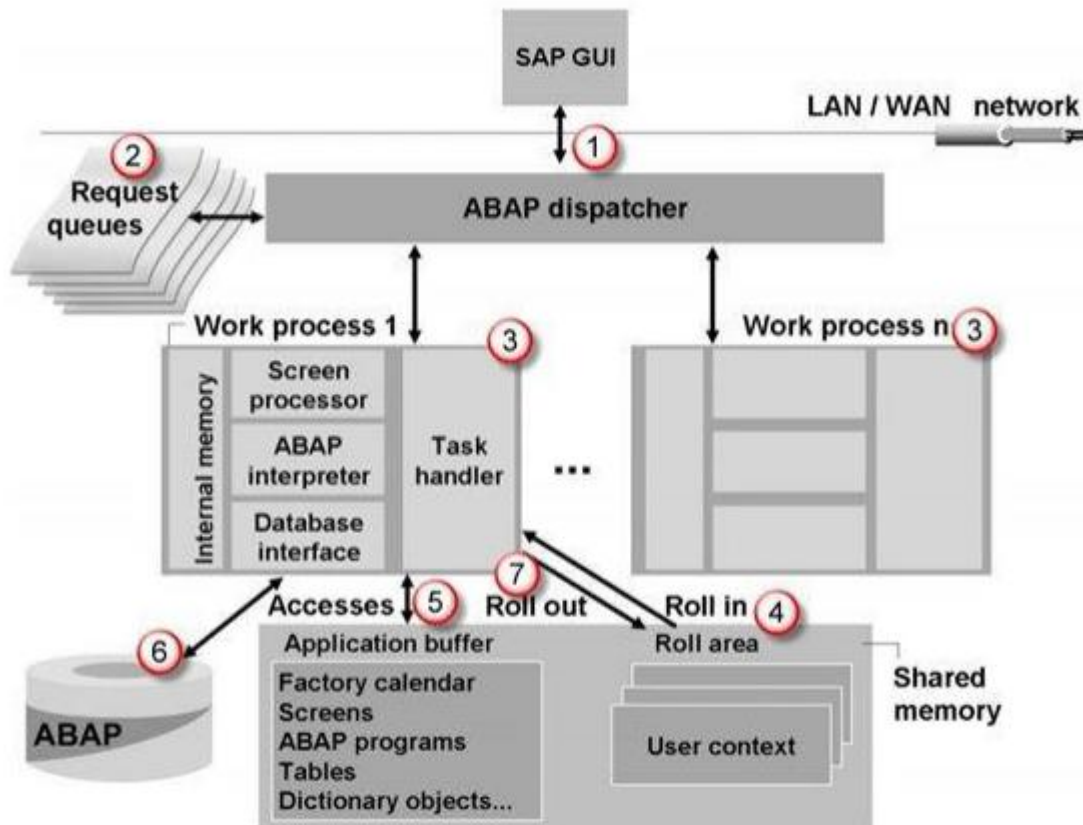
12. Java Server Processes: It can process a large number of requests simultaneously.

13. Threading: Multiple Processes executes separately in the background; this concept is called threading.

14. ICM: It enables communication between SAP system and HTTP, HTTPS, SMTP protocol. It means by entering system URL in the browser you can access SAP from browser also.

One more component is JCO. JCO is used to handle communication between java dispatcher and ABAP dispatcher when system is configured as ABAP+Java.

SAP Logon Process Work



Step 1) Once a user clicks on the SAP system from GUI, the user request is forwarded to **Dispatcher**.

Step 2) Request is stored in **Request queues first**. Dispatcher follows **First in First out rule**. It will find free work process and if available will be assigned.

Step 3) As per user request, particular work process is assigned to user. For example, when user login to the system then Dialog work process is assigned to the user. If user runs a report in background, then background work process is assigned to the user. When some modifications are done at database level then update work process is assigned. So as per user's action work process is assigned.

Step 4) Once user is assigned the dialog work process then user authorizations, user's current setting are rolled in to work-process in shared memory to access user's data. Once dialog step is executed then user's data is rolled out from work process. Thus shared memory will be cleaned and other user's data can be saved in shared memory area. Dialog step means the screen movements. In a transaction, when a user's jumps from one screen to other the process is called a dialog step.

Step 5) First work process will find the data in the buffer. If it finds data in buffer, then there is no need to retrieve data from database. Thus response time is improved and this process is called hit. If it does not find the data in buffer, then it will find the data in database and this process is called miss. Hit ratio should be always higher than miss ratio. It improves the performance of system.

Step 6) Other requested data is queried from the database and once the process is complete, the result is sent back to GUI via dispatcher.

Step 7) At the end user's data is removed from shared memory so the memory will be available to other users. This process is called rollout.

Client Creation

What is a SAP Client

Client is a 'Customer'. in SAP. We can say that each customer maps to one client. Within one SAP instance, a number of Clients can be created. No need to install separate software's for each and every customer. It provides isolation, one client cannot see the data of another client.



As depicted above 100 and 200 clients exist under one roof. We can create a number of clients in SAP Application (from 000 to 999).

Advantages of Client Concept

Client concept comes in with the following advantages

- You can share the same resources between multiple users.
- You can manage SAP system landscape as you can create multiple clients for DEV, QA and PROD team.
- You can share your SAP system with a large number of users.
- You can create clients in SAP system from **000-999**.

What does client contain

1. Application Data - Application data is the data that are stored in the database tables.

2. Customizing Data - Customizing data is data created by customers when they customize their systems

3. User Master Record - A user master record defines the authorizations assigned to a user. Basis consultants are responsible for maintaining the user master record and assigning authorizations.

Advantages of Client concept

1. Clients enable SAP SAS providers to install a small number of SAP Systems, but still cater to a large number of customers.
2. Costs are not only saved by sharing hardware and software but multiple customers also use the same application solution, including administration and support.
3. Clients help establish your SAP landscape. For instance, you can have a client for the development team, a client for a test team and a production client.

SAP comes with three "standard clients":

1. 000
2. 001
3. 066

000 Client: - We can find this client in the system as soon as we install SAP r/3 software. This is called master client. Client 000 contains a simple organizational structure of a test company and includes parameters for all applications, standard settings, and configurations for the control of standard transactions and examples to be used in many different profiles of the business applications. It contains client independent data.

001 Client: - This client is a copy of the 000 client including the test company. This client's settings are client-independent if it is configured or customized. People normally use 001 clients to create a new client.

066 Client: - This client is called early watch client. The SAP early watch alert is a diagnosis service, for solution monitoring of SAP and non-SAP systems in the SAP Solution Manager. Alert may contain Performance issue, average response time, current system load, Database administration, etc.

How to create a new Client

Theoretically, we can create clients from 000 to 999. But maintenance of such a large number of clients becomes a challenge.

Step 1) Execute T-Code **SCC4**



Step 2) It will bring you to the initial screen of SAP clients.

Display View "Clients": Overview

Client	Name	City	Crcy	Changed on
000	SAP AG Konzern	Walldorf	EUR	22.03.2011
066	Test EarlyWatch Profiles	Walldorf	EUR	09.05.2003

Click New Entry to make a new SAP Client

Display View "Clients": Overview

Client	Name	City	Crcy	Changed on
000	SAP AG Konzern	Walldorf	EUR	22.03.2011
066	Test EarlyWatch Profiles	Walldorf	EUR	09.05.2003

Step 3)

1. Enter basic details as given below.
 - Client number & description
 - City to which client Belongs (etc, NY-New York)
 - Logical system may be <Std>CLNT<Client Number>
 - Std Currency may be (etc EUR)
 - Client roles may be Customizing, Demo, Training/Education, Production, etc..
2. Enter your client specific data and set permission for the clients as per your requirement
3. Save
4. Press F3 to come back to **SCC4**

Change View "Clients": Details

Client: 100 | Dev Client

City: Amsterdam

Logical system: SD1CLNT100

Std currency: EUR

Client role: Customizing

Last Changed By: DDIC

Date: 22.03.2011

Client specifications

Changes and Transports for Client-Specific Objects

☐ Changes without automatic recording

☒ Automatic recording of changes

☐ No changes allowed

☐ Changes w/o automatic recording, no transports allowed

Cross-Client Object Changes

Changes to Repository and cross-client Customizing allowed

Protection: Client Copier and Comparison Tool

Protection level 0: No restriction

CATT and eCATT Restrictions

eCATT and CATT Allowed

Restrictions

☐ Locked due to client copy

☐ Protection against SAP upgrade

Step 4) New client will be there in the list. Here we have created client 100.

Display View "Clients": Overview

Client	Name	City	Crcy	Changed on
000	SAP AG Client	Walldorf	EUR	22.03.2011
055	Test Client	Walldorf	EUR	09.05.2003
100	Dev Client	Amsterdam	EUR	22.03.2011

Client Copy

Client Copy

We can generate a blank client with SCC4. **But how to fill the data in the client? "Answer is the client copy."**

Client copy means **"transferring client specific data"** within the same instance(SID) or between different instances(SID).

Client copy can be performed with three different methods

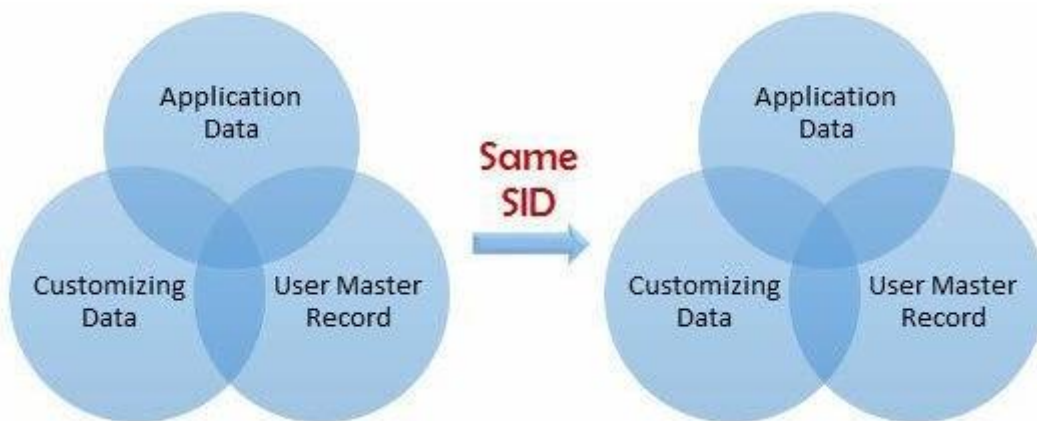
1. Local client copy.

2. Remote client copy.

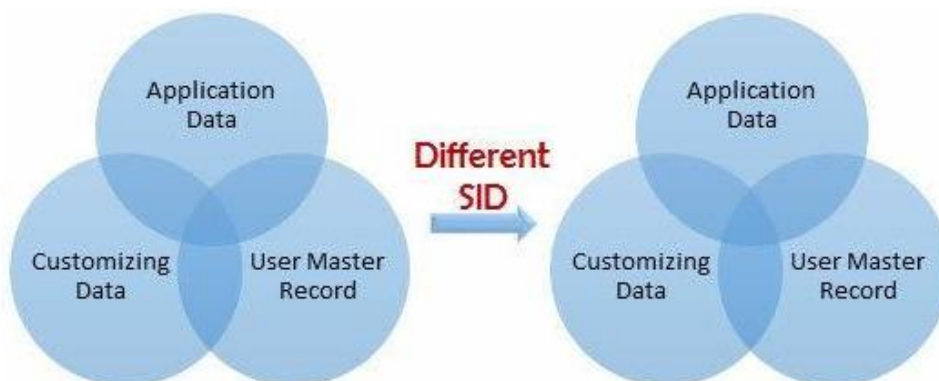
3. Client Import/Export.

Below brief details are given about client copy methods.

Local Client Copy: This method is used to copy client within the same instance (SID). It is done by T-code **SCCL**.



Remote Client Copy: This method is used to copy client between different instances(SID). It is performed by T-code **SCC9**.

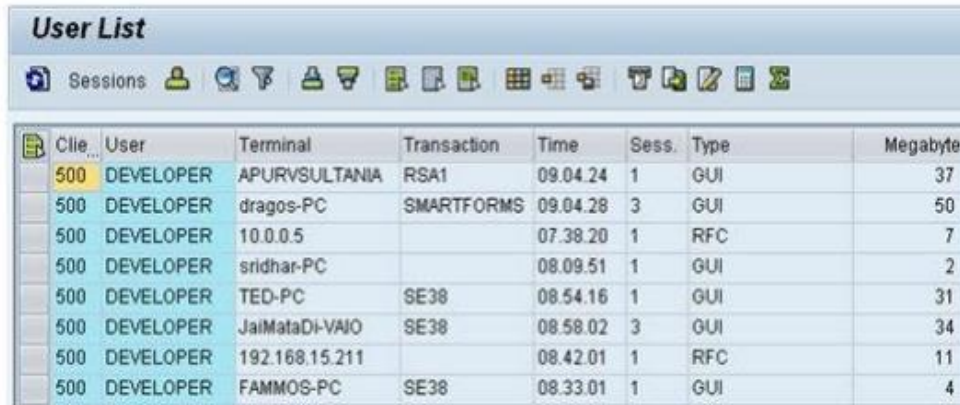


Client Import/Export: This method is used to copy client between different instances(SID). It is performed by T-code **SCC7/SCC8**.

Client Copy Pre-steps

To avoid data **inconsistencies** there are few pre-steps to be performed before starting client copy:

- 1) **Disconnect and lock business users(SU10).** You can end the session of active users in the system through **SM04**. Once all users are logged out, check that no cancelled or pending update requests exist in the system.



The screenshot shows the 'User List' table in SAP SM04. The table has columns: Clie, User, Terminal, Transaction, Time, Sess., Type, and Megabyte. It lists several active sessions for the user 'DEVELOPER'.

Clie	User	Terminal	Transaction	Time	Sess.	Type	Megabyte
500	DEVELOPER	APURVSULTANIA	RSA1	09.04.24	1	GUI	37
500	DEVELOPER	dragos-PC	SMARTFORMS	09.04.28	3	GUI	50
500	DEVELOPER	10.0.0.5		07.38.20	1	RFC	7
500	DEVELOPER	sridhar-PC		08.09.51	1	GUI	2
500	DEVELOPER	TED-PC	SE38	08.54.16	1	GUI	31
500	DEVELOPER	JaiMataDi-VAIO	SE38	08.58.02	3	GUI	34
500	DEVELOPER	192.168.15.211		08.42.01	1	RFC	11
500	DEVELOPER	FAMMOS-PC	SE38	08.33.01	1	GUI	4

2) Suspend all background jobs

- Execute **SE38** as given below.

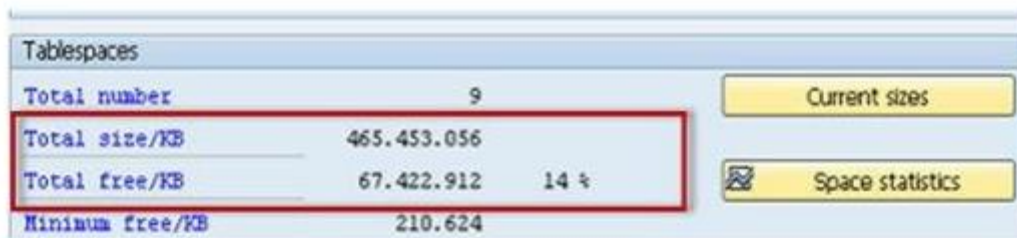


- Fill program name with **"BTCTRS1"** as above figure.
- Press Execute.



3) For a local copy, the system must have enough space in the database or table space.

For remote copy, target system must have enough space in the database or table space. Check space using Tx **DB02**.



Tablespaces			
Total number	9		
Total size/KB	465.453.056		
Total free/KB	67.422.912	14 %	
Minimum free/KB	210.624		

Buttons: Current sizes, Space statistics

4) To avoid inconsistencies during client copy users should not be allowed to work in source client.

5) **rdisp/max_wprun_time** parameter should be changed to 2000 second as a SAP recommendation. Although you use parallel processes and schedule job in the background, dialog processes will be used.

Local Client Copy

Local client copy is performed using Tcode **SCCL**.

Note: Here we have to Login Target Client.

Eg: Source Client (Default Client): 800

Target Client (Where we need to Transfer): 103

Pre-Requirements

- Check out whether Source Client & Target Client Existing or not.
- If not Create Client using T-code as SCC4
- We have to Locked all Active Users in Source Client
- **Check out Size of Source Client & Target Client by using the T-code as SE38, Program as SPACECHECK**
- We have to **Pause all Background Job's by using T-code as SE38**

Scenario:

- Source Instance & client: = DKM-000
- Target Instance & client: = DKM-202

Step 1) Create an entry for your new target client using **SCC4**. In our scenario, we will create client 202 in DKM system. Log on to this newly created target client (**DKM-202**) with user SAP* and default password pass.

Step 2) Execute T-code **SCCL**.



Step 3)

- Select your desired profile
- Enter Source client.
- Enter Description

Client Copy - Copy a Client

Schedule as Background Job Start Immediately

Target Client 100 Dev Client

Selected Profile SAP_CUST

Description Customizing

Source Client 000 SAP AG Konzern

☐ Test Run

Step 4) By default Client Copy is executed as a single process. A Single process will take a lot of time. We will distribute the workload of single the process to parallel(multiple) processes which will reduce time in copying a client.

1. Select **GoTo** from the menu bar.
2. Select **Parallel Process**. Parallel processes are used to exploit the capacity of database better

Client Copy -

GoTo Profile System Help

Request Copy F7

Delete Client F9

Log Display

Parallel Processes Shift+F1

RFC Destinations

Back F3

Step 5) Always execute long running processes in background mode rather than foreground/dialog mode. In fact, some processes run more quickly in the background.

Client Copy - Copy a Client

Schedule as Background Job Start Immediately RFC System Comparison

Target Client 100 Dev Client

Selected Profile SAP_ALL

Description All Client-Specific Data w/o Change Documents

Source destinat. BD1CLNT101

System Name BD1

Source Client 101

☐ Test Run

Step 6) The client copy logs are available in **SCC3**. Status - "**Successfully Completed**" means client copy is completed.

Date	Time	Source	Status Text	Profile	Mode	Test
04.09.2006	18:03:29	200	Successfully Completed	SAP_ALL	Local	
23.08.2006	01:00:27	200	Successfully Completed	SAP_ALL	Local	
18.07.2006	08:28:46	000	Successfully Completed	SAP_UCSV	Local	
17.07.2006	07:45:04	200	Successfully Completed	SAP_USER	Local	

Remote Client Copy

This technique uses Remote function call. You can view RFC from SM59. This technique depends on the network, so network connectivity must be strong enough.

Note: Logon with Target Client

Eg: Source Client 800 Target Client 101

Pre-requisites:

- For this we require Logical System & RFC
- 1st we need to check whether client having Logical System or not (In Source Client). If not, we have to create Logical System with the T-code as **BD54**
- 2nd we need to check whether client having RFC or not. If not, we have to create the RFC with T-code as **SM59** (In Source Client)

Scenario:

- Source Instance & client: = **BD1-101**
- Target Instance & client: = **DKM-202**

Step 1) Log on to the target system. Here we will log on to DKM system. Create a new target client entry (202) using **SCC4**. Log on to this new target client with user **SAP*** and default password "**pass**". Here we will log on to DKM-200 system.

Step 2) Execute Transaction Code **SCC9**.



Step 3) Fill the basic details as per your requirement.

Client Copy - Copy a Client

Target Client: 100 Dev Client
 Selected Profile: SAP_ALL
 Description: All Client-Specific Data w/o Change Documents
 Source destinat.: BD1CLNT101
 System Name: BD1
 Source Client: 101
☐ Test Run

Step 4) Select Parallel Process. Parallel processes are used to exploit the capacity of database better.

Client Copy - Client

1 Goto Profile System Help

2 Parallel Processes Shift+F1

Target Client

Selected Profile

Request Copy F7

Delete Client F9

Log Display

RFC Destinations

Back F3

Step 5) Schedule the client copy in background

Client Copy - Copy a Client

1

Target Client: 100 Dev Client

Selected Profile: SAP_ALL

Description: All Client-Specific Data w/o Change Documents

Source destinat.: BD1CLNT101

System Name: BD1

Source Client: 101

☐ Test Run

Step 6) The client copy logs are available in SCC3 as given below.

Client Copy/Transport Log Analysis

Client Copies in Client 202 : 5

Date	Time	Source	Status Text	Profile	Mode	Test
04.09.2006	18:03:29	200	Successfully Completed	SAP_ALL	Local	
23.08.2006	01:00:27	200	Successfully Completed	SAP_ALL	Local	
18.07.2006	08:28:48	000	Successfully Completed	SAP_UCSV	Local	
17.07.2006	07:45:04	200	Successfully Completed	SAP_USER	Local	

Client Import/Export

For large database, it is recommended to use client import/export instead of remote client copy.

Scenario:

Source Instance & client: = PKT-300

Target Instance & client: = DKM-202

This technique always starts with client **export** step.

Note: You must have enough space in the `/usr/sap/trans_SID` file system to perform the client export.

In SAP command line enter T-code as **SCC8/ SCC7 (Client Export/Import)**

How to export client

Step 1) Log on to the target system(DKM). Create an entry for your new target client using **SCC4**. Log on to the source system/source client(PKT).

Step 2) **Before you import a Client you need to export.** Export is nothing but transferring data files and co-files from source system's database to target system's import buffer. Execute T-code **SCC8**.



Step 3)

- Select profile
- Choose target system.



Step 4) Schedule the export in background



Step 5) Once the job is executed data files and co-files of profiles from **PKT system's database are transferred to DKM** system's import buffer. Once we will import request in DKM only then it will be reflected in a database of **DKM** system.

Depending on the chosen export profile there can be up to 3 transport requests created:

- Request PKTK000151 will hold the cross-client data,
- Request PKTKT00151 will hold the client dependent data,
- Request PKTKX00151 will also hold some client dependent data.

How to import the Client

Step 1) Log on to the newly created target client(DKM-202) using SAP* and password pass.

Step 2) Start the **STMS_IMPORT** transaction



As shown below, import queue will open

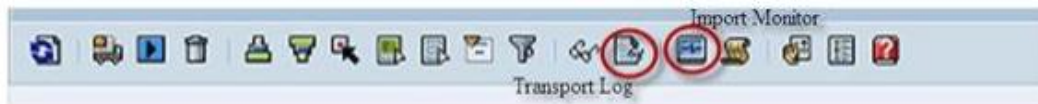
Import Queue: System N4S							
Requests for N4S: 1 / 29							
Request							
Number	Request	Clt	RC	Owner	Project	Short Text	St
1	E34K975388	500	Δ	RALPH	E34_P00011	TP_UPLOAD: Complete Delivery (Version 1.09 of 18	✓
2	E34K975784	500	Δ	VOLKER	E34_P00022	I62 Rollen für N4S - 08.11.2012	✓
3	E34K975596	800	Δ	VOLKER	E34_P00025	IDES Entwicklungen - 2.32 - complete	✓
4	E34K975558	800	Δ	VOLKER	E34_P00020	System-Check W0 V02.36 - 13.09.2012 - Complete	✓
5	E34K975768	500	Δ	VOLKER	E34_P00022	Z162_EXT_PORTAL	✓
6	E34K973239	500	Δ	RALPH	E34_P00036	EXPLORER: Standardlayout	✓
7	E34K973259	500	Δ	RALPH	E34_P00036	EXPLORER: Auslieferungscustomizing (Stand 01.04.2	✓
8	E34K973803	500	Δ	RALPH	E34_P00036	EXPLORER: Auslieferung Rolle (03.02.2011)	✓
9	E34K975348	500	Δ	RALPH	E34_P00036	EXPLORER: Complete Delivery (Version 1.08 of 28.0	✓
10	E34K974191	500	Δ	RALPH	E34_P00036	EXPLORER: Sicherheitsprüfung aktivieren (Version	✓
11	E34K974193	500	Δ	RALPH	E34_P00036	EXPLORER: Sicherheitsprüfung deaktivieren (Versio	✓
12	E34K975764	800	Δ	VOLKER	E34_P00022	I62 Rollen für N4S - 08.11.2012	✓
13	E34K975768	800	Δ	VOLKER	E34_P00022	Z162_EXT_PORTAL	✓
14	E34K974129	800	Δ	VOLKER	E34_P00025	IDES Entwicklungen - 2.08	✓

Step 2) Select the transport requests generated by client export. Import these transport requests on the target client.

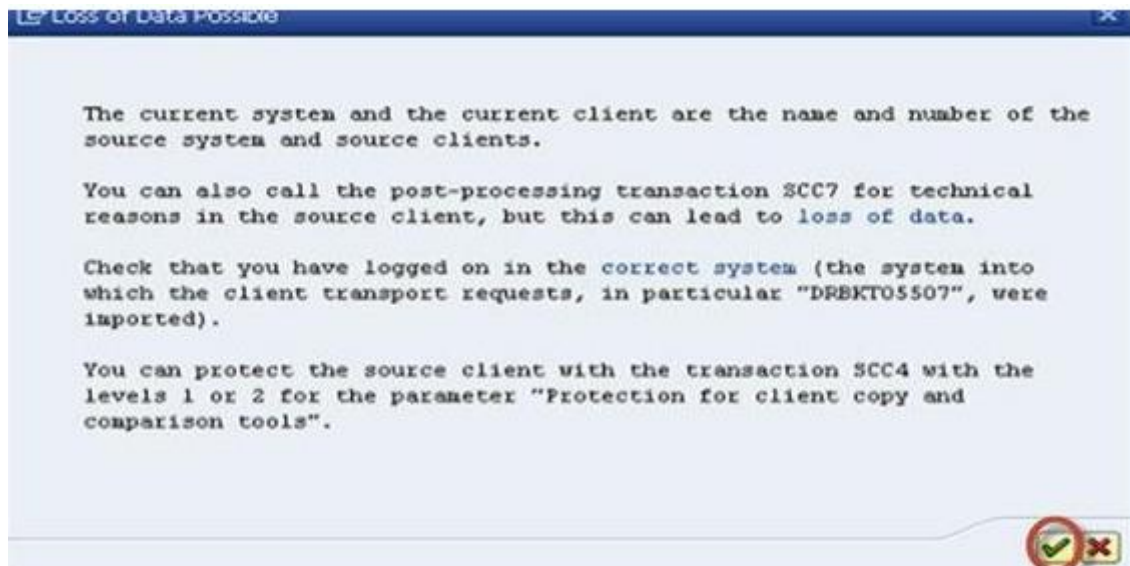
The transport requests should be imported in the following sequence:

1. Request PKTK000151
2. Request PKTKT00151
3. Request PKTKX00151 The system automatically detects these are client export transport requests and automatically performs the import of the 3 requests.

The import logs can be seen in **STMS_IMPORT**.



Step 3) Post import phase: - Once the import is done, execute **SCC7** to perform the post client import actions,



Schedule the post import job in background.



Step 4) Import log will be available in **SCC3**. The Client is successfully imported.

Client Deletion

Following are the detailed steps to Delete a client in SAP

Step 1) T-code which is used for client deletion is **SCC5**.



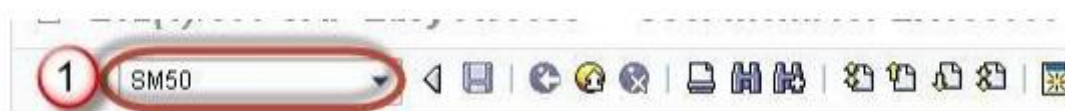
Step 2) Click on "delete in the background" to run client deletion as background job. You can also check the option "Delete an entry from T000" table.



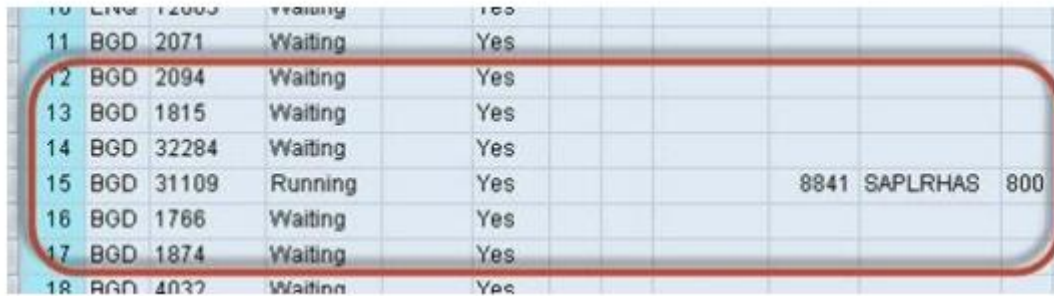
Table "T000" contains clients' entry which we have created in **SCC4**.

Data Browser: Table T000 Select Entries 5				
Check Table...				
Table: T000				
Displayed Fields: 17 of 17 Fixed Columns: 1				
MANDT	MTEXT	ORT01	MWAER	ADRN
000	SAP AG Konzern	Walldorf	EUR	
066	Test EarlyWatch Profiles	Walldorf	EUR	
500	MiniSAP external	Mannheim	EUR	
800	MiniSAP internal	Mannheim	EUR	

Step 3) Check the status of client deletion process using **SM50**.



Work process overview will open. "BGD" denotes background work process.



10	ENR	12000	Waiting	Yes				
11	BGD	2071	Waiting	Yes				
12	BGD	2094	Waiting	Yes				
13	BGD	1815	Waiting	Yes				
14	BGD	32284	Waiting	Yes				
15	BGD	31109	Running	Yes		8841	SAPLRHAS	800
16	BGD	1766	Waiting	Yes				
17	BGD	1874	Waiting	Yes				
18	BGD	4032	Waiting	Yes				

Once complete. Client will be deleted

Create a New User

Following are the detailed steps to Create a user in SAP

Step 1) Execute T-code SU01



Step 2)

1. Enter **Username** which you want to create.
2. Click the create button



Step 3) In the next screen

1. Click the Address tab.
2. Enter Details

Maintain User

User: DEMO
 Last Changed On: 00:00:00
 Status: Not saved

1 Address Logon data SNC Defaults Parameters Systems Roles

2

Person

Title: Mr.
 Last name: surname
 First name: XYZ
 Academic Title:
 Format:
 Function:
 Department: business
 Room Number: PCV-234 Floor: Building:

Communication

Language: English
 Telephone: Extension:
 Mobile Phone: 123456789
 Fax: Extension:
 E-Mail: XYZ@domain.com
 Comm. Meth: Remote Mail

Assign other company address... Assign new company address...

Step 4) Choose the user type in Logon Data tab.

There are 5 different user types available in SAP system.

1 Communication

2 Dialog

3 Reference

4 Service

5 System

Communication Users

Users are not allowed/possible to logon using SAP GUI and can't able to access the data through SAP GUI screens. Users are allowed to change the password if expired using external link.

SAP system always checks for the password expiry, initial password and prompts when it requires to change depends on the logon method (interactive or non-interactive).

These users main purpose is to use for external RFC calls. These users can access the SAP system data through frontend or external applications.

This user is used for external RFC calls.

Dialog Users

Users are allowed to logon using SAP GUI and can able to access the data through SAP GUI screens. System validates the password expiration, initial password and multi-logons.

The user can change the password by their own using SAP GUI. These users are individual users and individual personalized system access allowed.

This user is used for interactive system access from GUI.

Reference Users

Users are not allowed/possible to logon using SAP GUI and can't able to access the data through SAP GUI screens. System doesn't check for initial password and password expiration.

Users for general, non-person related which allows the assignment of additional authorizations to the users like internet users. These users used to give authorization to other users.

It is not possible to log on to the system with this user type. User type for general, non-person related users that allows the assignment of additional authorizations.

Service Users

Users are allowed to logon using SAP GUI and can able to access the data through SAP GUI screens. System doesn't check for initial password and password expiration.

Multiple logins are allowed for this kind of users. Users are not allowed to change the password and only administrator can change the password. These users have very restricted/minimum authorizations.

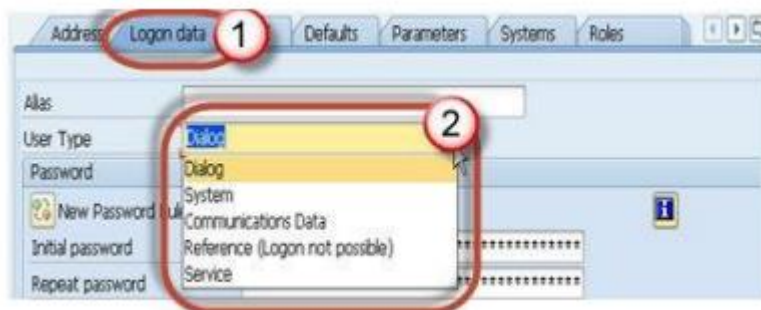
There type users are created for anonymous users. After an individual authentication, an anonymous session begin with service user can be continued as person-related session with a dialog user.

This user is created for a larger and anonymous group of users.

System Users

Users are not allowed/possible to logon using SAP GUI and can't able to access the data through SAP GUI screens. System doesn't check for initial password and expiration of password. Users are related System-related and internal system processes. The password change requirement does not apply to these user's passwords and it cannot be initial or expired. Only user administrator can change the password and multiple logons are permissible. These users are for background processing, external and internal RFC calls.

This user is used for background processing, communication within a system.



Step 5) Type the **initial password** for 2 times.

On first logon of the new user, system will ask to re-set the password.

Address Logon data **1** NC Defaults Parameters Systems Roles

Alias

User Type Dialog

Password

New Password Rules (Case-sensitive) **2**

Initial password

Repeat password

Password status

User Group for Authorization Check

User group

Validity Period

Valid from

Valid through

Other Data

Accounting Number

Cost center

Step 6)

1. Select the roles tab
2. Assign roles as per requirements

Defaults Parameters **Roles 1** Profiles Groups Personalization Licence...

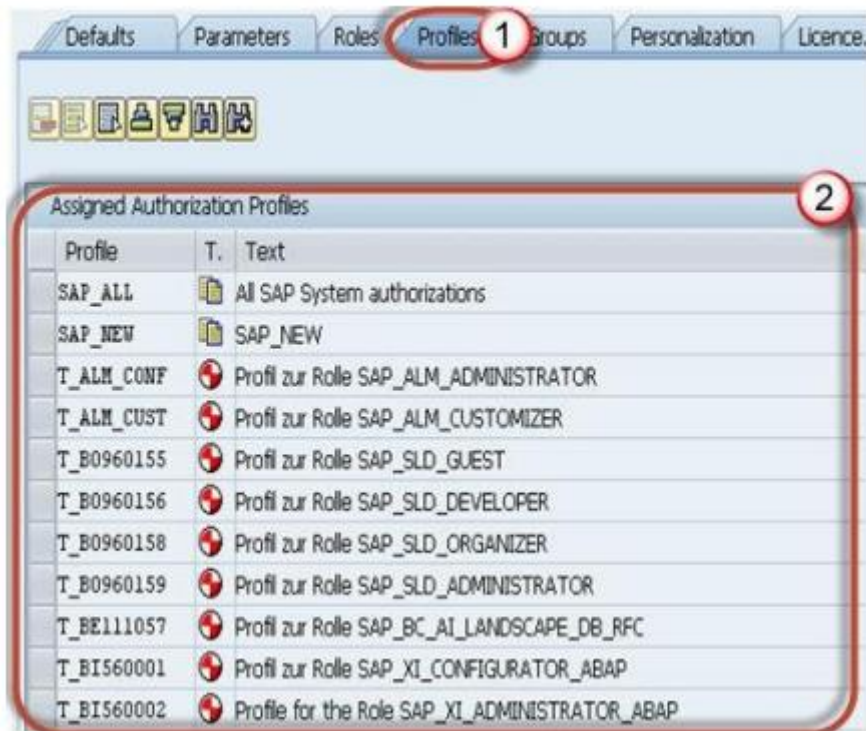
Reference user for additional rights

Role Assignments **2**

S..	Role	Ty...	Valid From	Valid to	Name
☒	SAP_ALM_ADMINISTRATOR		05.02.2013	31.12.9999	Alert Management Adminis
☒	SAP_ALM_CUSTOMIZER		05.02.2013	31.12.9999	Alert Management: Role fc
☒	SAP_BC_AI_LANDSCAPE_DB_RFC		05.02.2013	31.12.9999	Application Integration: RF
☒	SAP_SLD_ADMINISTRATOR		05.02.2013	31.12.9999	Rolle für die Administration
☒	SAP_SLD_CONFIGURATOR		05.02.2013	31.12.9999	Rolle für die Konfiguration
☒	SAP_SLD_DEVELOPER		05.02.2013	31.12.9999	Rolle für die Nutzung des S
☒	SAP_SLD_GUEST		05.02.2013	31.12.9999	Rolle für das Anzeigen von
☒	SAP_SLD_ORGANIZER		05.02.2013	31.12.9999	Rolle für die Organisation a
☒	SAP_XI_ADMINISTRATOR		05.02.2013	31.12.9999	Exchange Infrastructure: A
☒	SAP_XI_ADMINISTRATOR_ABAP		05.02.2013	31.12.9999	Exchange Infrastructure: A
☒	SAP_XI_ADMINISTRATOR_J2EE		05.02.2013	31.12.9999	Exchange Infrastructure: A

Step 7)

1. Select the **profiles** tab
2. Assign profiles as per requirements



You can assign **SAP_ALL** and **SAP_New** profile to user for full authorization.

- **SAP_ALL:** You assign this profile to users who are to have all R/3 authorizations, including super-user authorization.
- **SAP_NEW:** You assign this profile to users who have access to all currently unprotected components. The SAP_NEW profile grants unrestricted access to all existing functions for which additional authorization checks have been introduced. Users can therefore continue to work uninterrupted with functions which are subject to new authorization checks which were not previously executed.

Step 8)

1. Press save
2. Then the **back button (F3)** button



User will be created!

Users Lock & Unlock

Locking a user

The Purpose of locking user is to temporarily deactivate the users so that they cannot longer access the system.

Users can be locked in 2 ways:

- Automatically
- Explicitly/Forcefully

Automatically: There are two possibilities when users get lock automatically

- Maximum number of failed attempts: - controlled via the parameter login/fails_to_user_lock. If a value is set to 3 it means after 3 failed attempts user will be locked.
- Auto unlock time: - "login/failed_user_auto_unlock" defines whether user locked due to unsuccessful logon attempts should be automatically removed at midnight.

Explicitly/Forcefully: We can lock and unlock users in 2 ways

1. Lock single user (SU01)

2. Lock multiple user (SU10)

Procedure to lock a single user

Step 1) Execute T-code SU01



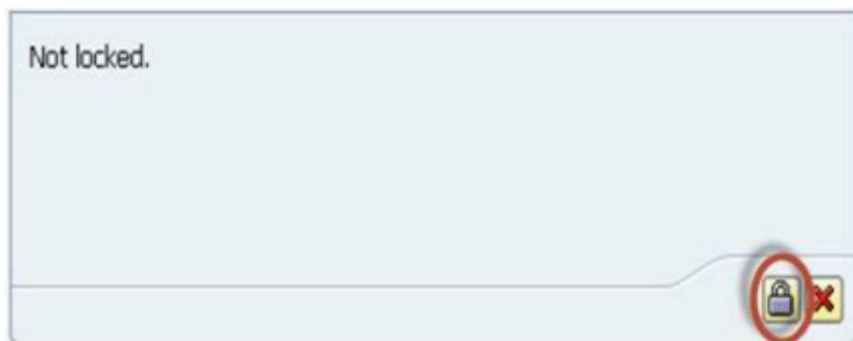
Step 2) Enter a username in **User** field.



Step 3) Press **Lock/Unlock** button



Step 4) In the next screen, Press **Lock** button again to lock the user.

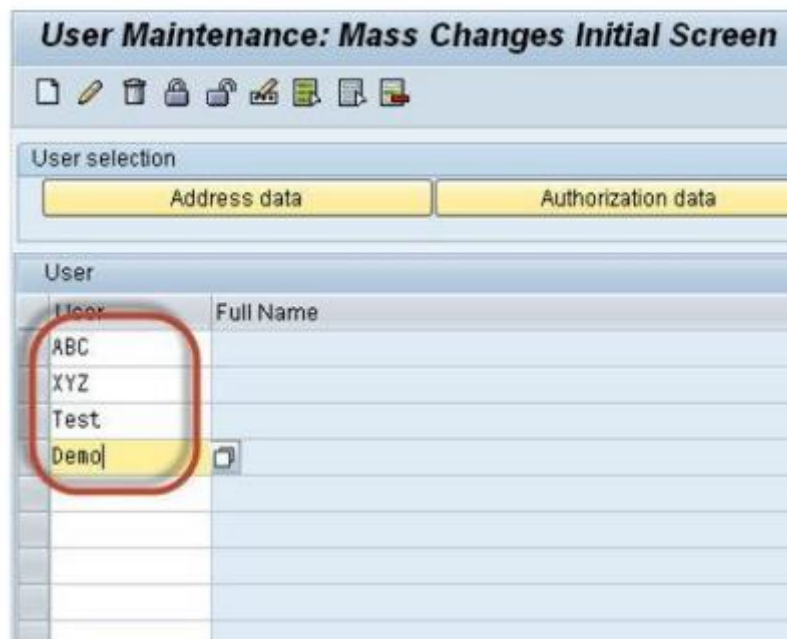


Procedure to lock multiple users

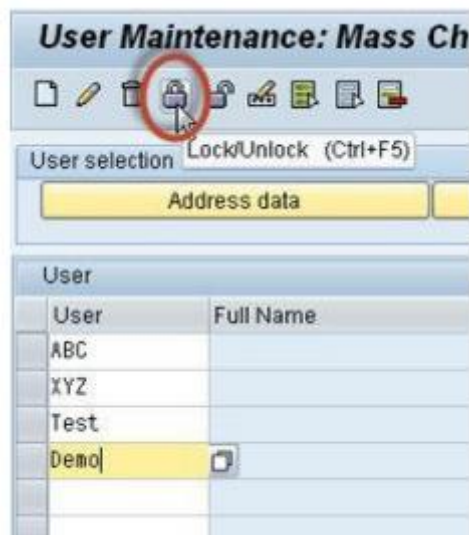
Step 1) Execute T-code SU10



Step 2) Enter users a username in **User** field.



Step 3) Press **Lock/Unlock** button



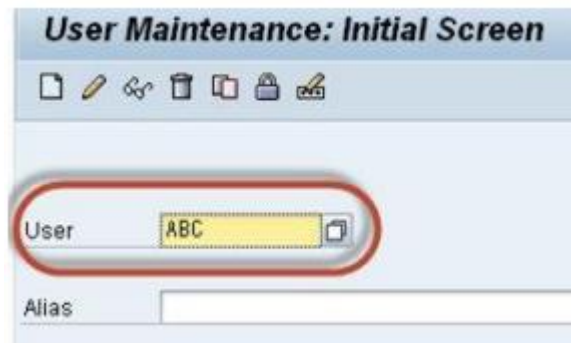
All the users listed will be locked

Procedure to unlock a user

Step 1) Execute T-code **su01**



Step 2) Enter username in **User** field.



Step 3) Press **Lock/Unlock** button



Step 4) Press **Unlock** button

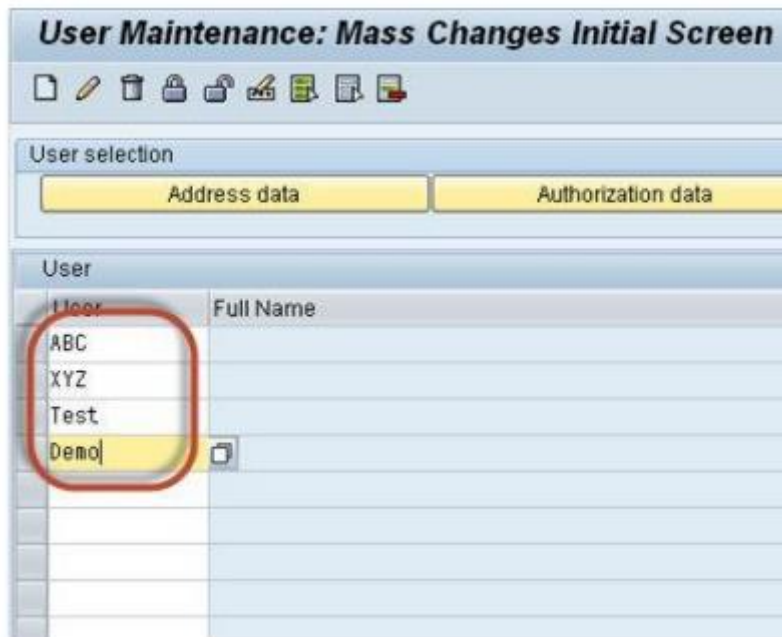


Procedure to unlock multiple users

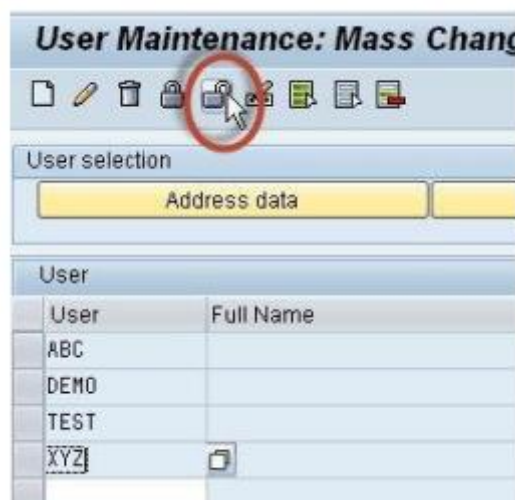
Step 1) Execute T-code **SU10**



Step 2) Enter users' username in **User** field.



Step 3) Press **Unlock** button



Users will be unlocked

Logon Limit Attempts

Before we learn to limit logon attempts we need to know parameter

What is a parameter

Parameter is the set of keys and values to manage the SAP system. There are two types of parameters

- 1. Static:** - It needs a restart. It doesn't effect to the system immediately once you set the value for it.
- 2. Dynamic:** - It does not need restart. It effects to the system immediately once you set the value for it.

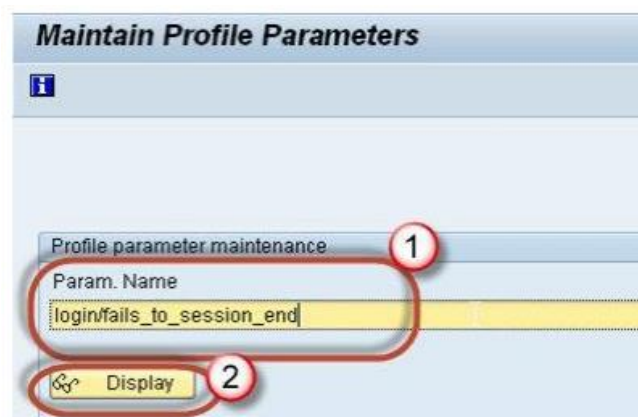
How to view a Parameter?

Step 1) Execute T-code **RZ11**.

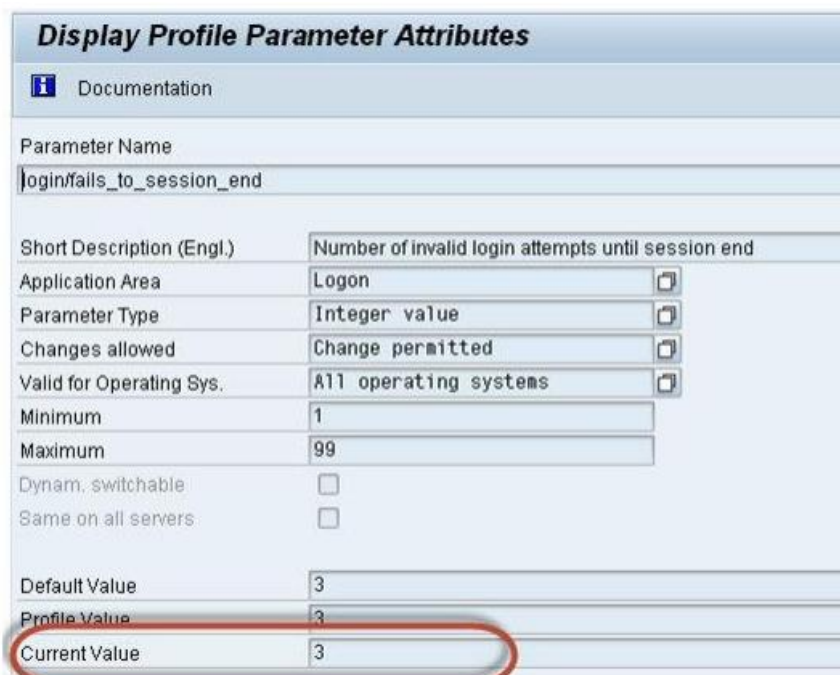


Step 2)

1. Put parameter name "**login/fails_to_session_end**" in text field. You can put any Parameter name.
2. Click Display



Step 3) The screen below shows the current value set for the parameter by the admin



In order to change a parameter, click the pencil icon and make desired changes.

Important Parameters to limit login attempts

- **login/fails_to_session_end:** This parameter specifies the number of times that a user can enter an incorrect password before the system ends the logon attempt. The parameter is to be set to a value lower than the value of parameter
- **login/fails_to_user_lock:** This parameter specifies the number of times that a user can enter an incorrect password before the system locks the user against further logon attempts. Default value is 12. You can set it to any value between 1 and 99 inclusive.

Incorrect Login Parameters

- **login/fails_to_session_end:** Specifies the number of unsuccessful logon attempts. The system does not allow any more logon attempts. The parameter set to be a value lower than the login/fails_to_user_lock value. The default value is 3. The allowed values are 1 to 99.
- **login/fails_to_user_lock:** Specifies the number of unsuccessful logons attempts before the system locks the user. The default value 5. The allowed values are 1 to 99.
- **login/failed_user_auto_unlock:** Specifies whether user locks due to unsuccessful logon attempts should be automatically removed at midnight. The default value is 0 (locks due to incorrect logon attempts remain in force for an unlimited period). The allowed values are 0 or 1.

Password Restrictions

Every user requires a configured userid and password to access the SAP system. Initially, the password configured by the SAP system administrator.

On the first logon, user needs to change the password according to the rules/restrictions configured in the system by administrator. The rules/restrictions are configured in the system with some set of the parameters. The parameters with password rules are called as password rules parameters.

Password rules parameters

Parameter	Description
login/min_password_lng	Specifies the minimum length of the password. The default value is 6. The allowed values are from 3 – 40.
login/min_password_digits	Specifies the minimum number of digits (0-9) in passwords. The default value is 0. The allowed values are from 0 – 40.
login/min_password_letters	Specifies the minimum number of letters (A-Z) in passwords. The default value is 0. The allowed values are from 0 – 40.
login/min_password_lowercase	Specifies the number of characters in lower-case letters a password must contain. The allowed values are from 0 – 40. The default value is 0.
login/min_password_uppercase	Specifies the number of characters in upper-case letters a password must contain. The allowed values are from 0 – 40. The default value is 0.
login/min_password_specials	Specifies the minimum number of special characters in the password. The special characters allowed are !"@ \$%&/()=?`*+~#-_.,:{}[]\<> and space and the grave accent. The default value is 0. The allowed values are from 0 – 40.

login/password_charset

This parameter specifies the password characters set. Allowed values are:

- 0 (restrictive): The password can consist only following (ASCII) special characters: !"@ \$%&/'()*=?*+~#-_.,:;{}[]\<> | and space and the grave accent.
- 1 (backward compatible, default value): The password can consist of any characters including national special characters (ISO Latin-1, 8859-1).
- 2 (not backward compatible): The password can consist of any characters.

There are some other set of parameters with rules which are in effect when changing the password in the SAP system. Those parameters are called as password changes parameters.

Password changes parameters

Parameter	Description
login/min_password_diff	Specifies the minimum number of characters that must be different in the new password compared to the old password. The default value is 1. The allowed values are from 1 – 40.
login/password_expiration_time	Specifies the validity period of passwords in days. The default value is 0. The allowed values are from 0 – 1000.
login/password_history_size	Specifies the number of passwords that the system stores and that the user cannot use again. The allowed values are from 1 – 100. The default value is 5. In this unit is number of entries.
login/password_change_waittime	Specifies the number of days that a user must wait before changing the password again. The allowed values are from 1 – 1,000. The default value is 1. In this unit is days.

Table USR40: Specifying Impermissible Passwords

Users can be prevented from choosing passwords that administrator/company do not want to allow. Table USR40 contains the prohibited rules for the users.

To add new restriction, the restriction should be entered in table USR40. To maintain the table USR40, SM30 transaction can be used. There are two wildcard characters:

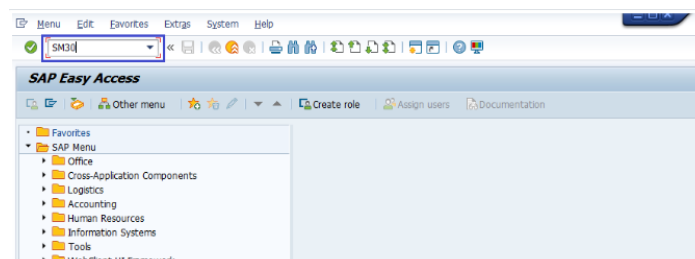
- ? - Specifies a single character
- * - Specifies a sequence of characters in any combination of any length.

Example

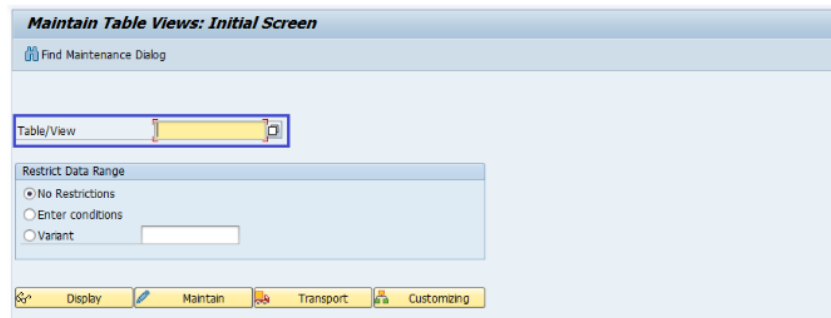
- 567* - Rejects any password that begins with the sequence “567”.
- *567* - Rejects any password that contains the sequence “567.”
- KL? - Rejects all passwords that begin with “KL” and have one additional character like “KLA”, “KLB”, “KLC” and so on.

Table USR40: Adding new restriction

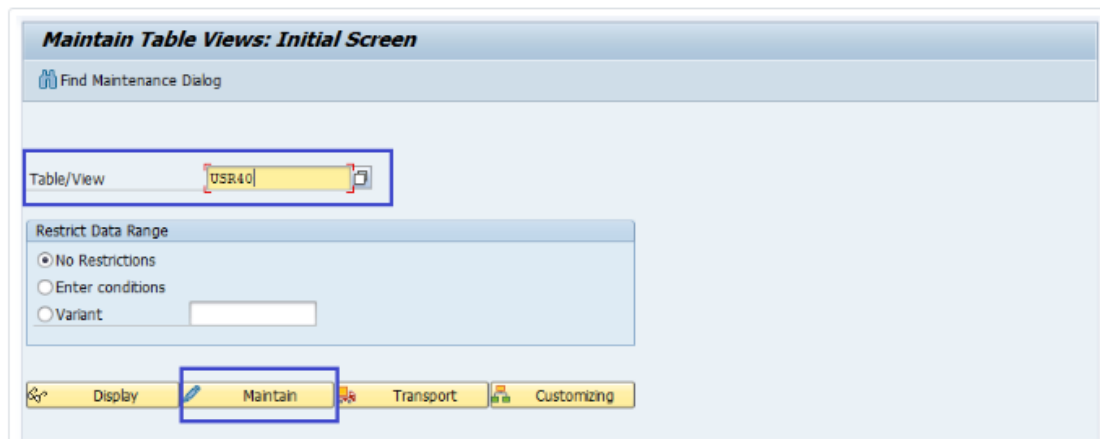
Step-1: Go to SM30.



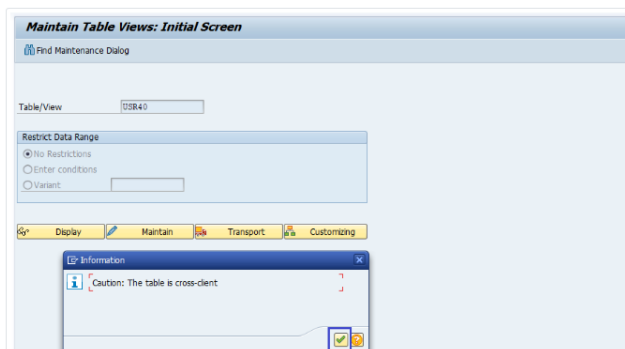
Step-2: It navigates to the “Maintain Table Views: Initial Screen”.



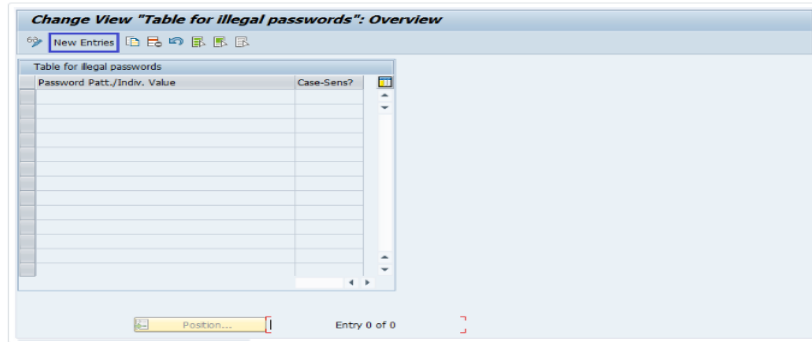
Step-3: Enter USR40 in the “Table/View” field. Click on the maintain icon.



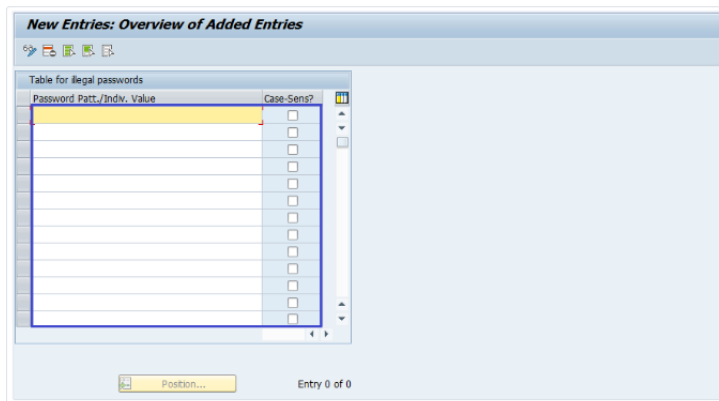
Step-4: It displays an informational dialog box showing a caution like below. Click on tick mark.



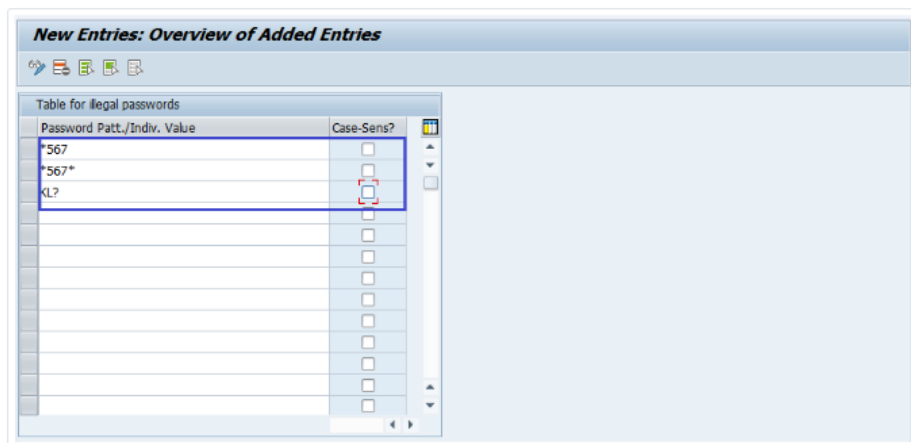
Step-5: Click on the new entries in the below screen to add a restriction.



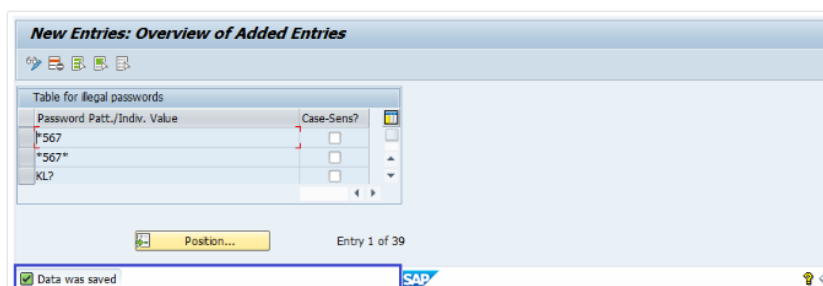
Step-6: Now the table is editable.



Enter the restrictions to the table below and click on save to restrictions become active.



Step-7: It prompts for a workbench request. Click on the right mark to proceed. Once the work bench request completed, all the restrictions added in effect.



Background Job Create, Schedule, Reschedule

What is a Background Job

Background job is a non-interactive process that runs behind the normal interactive operations. They run in parallel and do not disturb interactive (foreground jobs) processes and operations.

It is scheduled from **SM36**. You can analyse it from **SM37** by viewing its job log.

Advantages of Background Jobs

- It reduces manual effort & automates the task.
- It can be scheduled as per user's choice.
- It reduces user interaction and can run seamlessly in the background without user input
- Once you define the variant for background job, the user doesn't have to worry about value input in the field. Thus, user confusion is also reduced.
- Ideal for time- consuming/resource intensive programs which can be scheduled to run in the night (when system load is low).

Background jobs are classified into three categories

- 1. Class A (High/critical Priority):** - Some tasks are urgent or critical and must be scheduled with class A priority job. Class A priority reserves one or more background work processes. Users have to decide how many background work processes should be assigned to Class A priority job. Suppose a user chooses 2 background work processes for this category then available background work processes for class B and C = (Total number of work processes set in operation modes RZ03)- (Background work processes allowed to class A category).
- 2. Class B (Medium Priority):** - Once Class A jobs are completed, Class B job will start executing in the background before class C jobs.
- 3. Class C (Low Priority):** - It runs after both class A and class B jobs are completed.

Possible status of background jobs

- 1. Scheduled:** - You have defined the program name and variant but not defined start condition like Start Date, End Date, Frequency etc. That means you have not defined when a job should be scheduled in system.
- 2. Released:** - All required criteria are fulfilled for job definition. Start condition is must for the job to be in release status.
- 3. Ready:** - All the required conditions are met to run the job in a background work process. But job scheduler has put the job in the queue because it is waiting for background work process to be free.
- 4. Active:** - Job has started running in the background. We cannot change the status of the job once it is in Active status.
- 5. Finished:** - Job is executed successfully. It means the desired task is completed without any error.

6. Cancelled: - There are two possibilities for this. The Administrator has forcefully cancelled the job or there might be some issue with job. You can investigate this from Job logs.

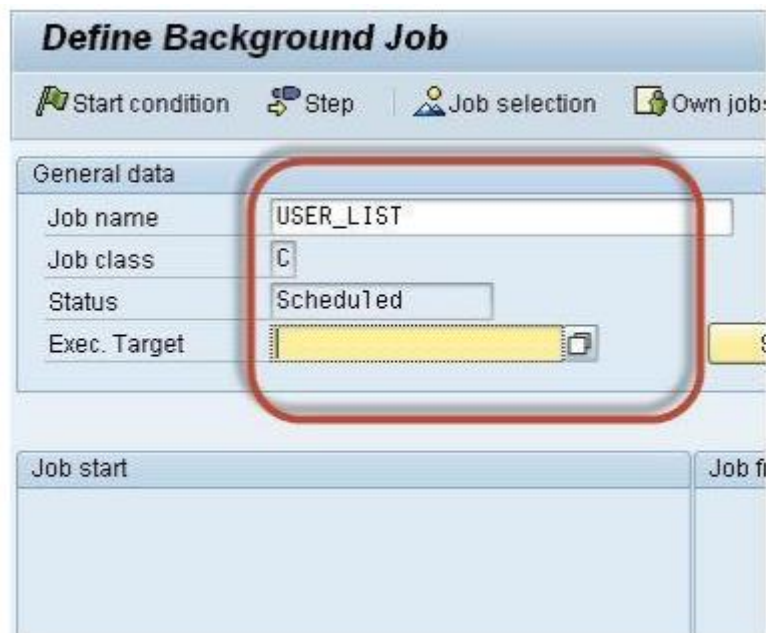
How to schedule the background job

You can schedule the background job using SM36. Planned or immediate jobs can be scheduled.

Step 1) Execute T-code SM36.



Step 2) Fill the job name, priority (A/B/C) and the target server. Background jobs once scheduled on a target server run on that server. Main purpose of defining target server is the workload balancing.



Step 3) Click on "spool list recipient". You will get output in your mailbox. You can check email from SBWP.



Step 4) Insert your SAP username and click the copy button.

N4S(3)/500 Recipient Determination

Recipient: DEVELOPER

General attributes:

☐ Copy ☐ Blind copy

☐ Express ☐ No forwarding

Buttons: Copy (checked), Fax entry, X.400 entry, Address, Close

Step 5) Click **Step** button to define ABAP program, variant's details, etc.

Define Background Job

Buttons: Start condition, Step (circled), Job selection, Own jobs

General data:

Job name: USER_LIST

Job class: C

Status: Scheduled

Exec. Target:

Define steps (F6)

Step 6) Define program name, variant details.

1. Enter your program name, Variant name in the field. If you have not created variant as per your requirement, then leave it blank.

2. Press save button.

Program values

Buttons: ABAP program, External command, External program

ABAP program:

Name: RSUSR200

Variant: M_TEST

Language: EN

External command (command pre-defined by system administrator):

Name:

Parameters:

Operating sys.:

Target server:

External program (direct command input by system administrator):

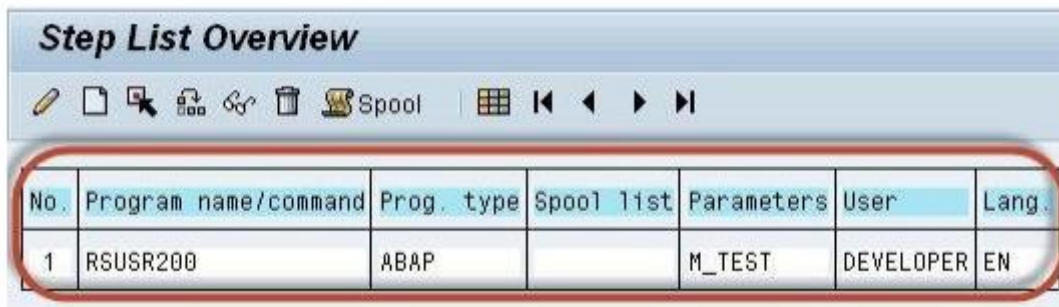
Name:

Parameter:

Target host:

Buttons: Check, Save (circled), Print specifications, Close

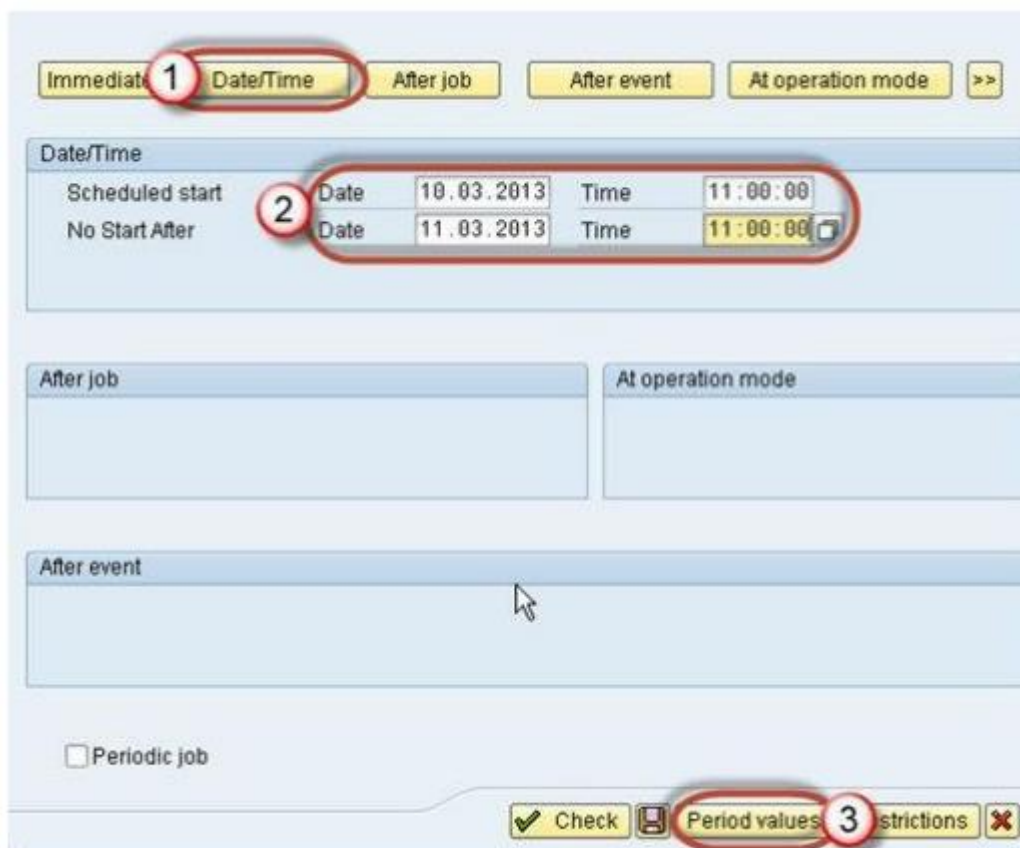
Step 7) Once you schedule the job you will get the following screen.



No.	Program name/command	Prog. type	Spool list	Parameters	User	Lang.
1	RSUSR200	ABAP		M_TEST	DEVELOPER	EN

Step 8) Click Start conditions to fill start date, end date, frequency, etc. for job. If you do not specify start condition, **then job will always remain in scheduled status.** A job in scheduled status will never run.

1. Click on Date/Time (For periodic jobs). If you click "Immediate" then job will start running right away. But it will not be set as periodic job. It's like "press and run."
2. Define job's start date/time, end date/time. The job will be released only once it meets its Scheduled start date/time.
3. Press periodic values.



Immediate **1** Date/Time After job After event At operation mode >>

Date/Time

Scheduled start	2 Date	10.03.2013	Time	11:00:00
No Start After	Date	11.03.2013	Time	11:00:00

After job At operation mode

After event

☐ Periodic job

Check Period values **3** strictions

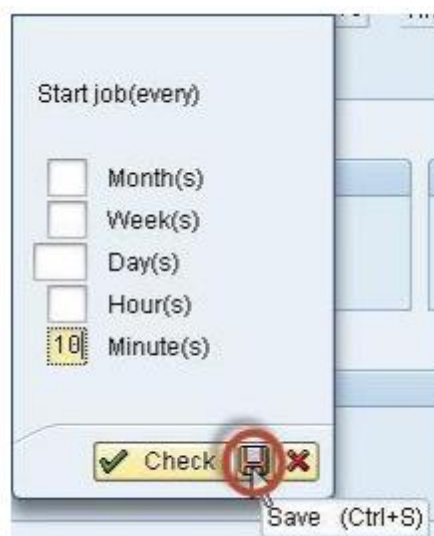
Step 9) Click on Hourly/Daily/Weekly period to define the frequency of the job as per your requirement. We will select Other Period



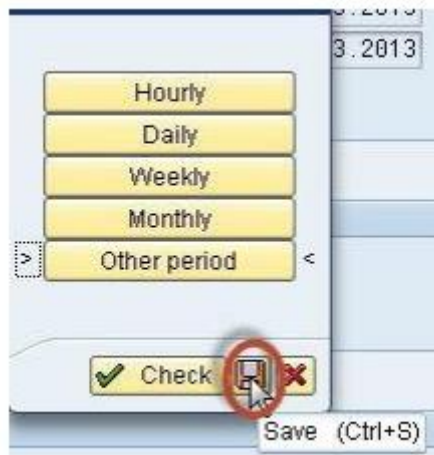
Step 10) Here you specify the recurring criteria of the job. For example, you can have the Job run after every 5 days from the Start Date. Here we select job to run every 10 minutes.



Step 11) Click on save button.



Step 12) Click on save again.



Step 13) Click save again

Step 14) Once **Job step and start conditions** are defined the following window will appear.

Define Background Job

Start condition Step Job selection Own jobs Job wizard

General data

Job name:

Job class:

Status:

Exec. Target:

Job start

Planned Start

Date: Time:

No Start after

Date: Time:

Job frequency

Job steps

1 Step(s) successfully defined

Job steps

Step 15) Press save.

✓ ◀ ▶ ↺ ↻ ⌂ ⌂ ⌂ ⌂ ⌂ ⌂

Define Background Job

Start condition Step Job selection Own jobs

General data

Job name:

Job class:

Status:

Exec. Target:

Step 16) GoTo **SM37** to know the status of the job.

Define Background Job

Start condition Step Job selection Own jobs Job wizard Standard

General data

Job name: USER_LIST

Job class: C

Status: Scheduled

Exec. Target: [Empty field]

Spool list recipient

Job start

Planned Start

Date: 10.03.2013 Time: 11:00:00

No Start after

Date: 11.03.2013 Time: 11:00:00

Job frequency

10 Minute(s)

Job steps

1 Step(s) successfully defined

Step 17) Select your criteria for the job which you want to monitor.

1. Put your job name and username who scheduled the job.
2. Select the status of the job.
3. Specify the date range. In our scenario, we just specify the end date while keeping from Date Open.